MIS 637 A DATA ANALYTICS & MACHINE LEARNING

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Step 1: Data Points

- We are provided with 10 two-dimensional points labeled from a to j. These points are to be grouped into 2 clusters using the k-means clustering algorithm. Below are the data points:
- a = (2, 0)
- b = (1, 2)
- c = (2, 2)
- d = (3, 2)
- e = (2, 3)
- f = (3, 3)
- g = (2, 4)
- h = (3, 4)
- i = (4, 4)
- j = (3, 5)

Step 2: Initial Cluster Centers

To begin k-means clustering with k=2, we choose two initial cluster centers as far apart as possible. We select point 'a' (2, 0), which is at the bottom-left of the point cloud, and point 'j' (3, 5), which is at the top-right.

Cluster 1 Initial Center: a = (2, 0)

Cluster 2 Initial Center: j = (3, 5)

Step 3: First Iteration - Cluster Assignment

Using the Euclidean distance formula, we calculate the distance of each point to the two centers. Each point is then assigned to the cluster whose center is closer. The distances are calculated as follows:

Euclidean Distance = $\sqrt{[(x2 - x1)^2 + (y2 - y1)^2]}$

Example: Distance between point b (1,2) and cluster center a (2,0) is $\sqrt{[(2-1)^2 + (0-2)^2]} = \sqrt{5} \approx 2.24$

After calculating all distances, we get:

Cluster 1: a, b, c, d

Cluster 2: e, f, g, h, i, j

Step 4: Compute New Cluster Centers

After assigning all points to clusters, we compute the new cluster centers by averaging the coordinates of points in each cluster.

Cluster 1 (a, b, c, d):
$\bar{x} = (2+1+2+3)/4 = 2$
$\bar{y} = (0+2+2+2)/4 = 1.5$
→ New Center = (2, 1.5)

Cluster 2 (e, f, g, h, i, j):
$\bar{x} = (2+3+2+3+4+3)/6 \approx 2.83$
$\bar{y} = (3+3+4+4+4+5)/6 \approx 3.83$
→ New Center ≈ (2.83, 3.83)

Step 5: Final Assignment After Convergence

After recomputing distances to the new centers and reassigning, the clusters remain stable. Therefore, the algorithm has converged with the final clusters being:

Cluster 1: a, b, c, d

Cluster 2: e, f, g, h, i, j

This result makes intuitive sense, as the points in Cluster 1 are positioned toward the bottom-left, while points in Cluster 2 are grouped toward the upper-right region of the plane.